


```
1 import os
2
3 %cd /content
4
5 if not os.path.exists('radiogenomics_ml'):
6     print("Clonazione di radiogenomics_ml")
7     !git clone https://github.com/andreagastaldi03/radiogenomics_ml
8 else:
9     print("radiogenomics_ml già presente, aggiorno i file")
10    %cd radiogenomics_ml
11    !git pull
12    %cd /content
13
14 %cd radiogenomics_ml
```

```
/content
radiogenomics_ml già presente, aggiorno i file
/content/radiogenomics_ml
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 4 (delta 3), reused 4 (delta 3), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.16 KiB | 295.00 KiB/s, done.
From https://github.com/andreagastaldi03/radiogenomics_ml
   21b6b62..0836aea  main      -> origin/main
Updating 21b6b62..0836aea
Fast-forward
 config.py          | 2 +-
 consensus_significance.py | 31 ++++++-----
 2 files changed, 26 insertions(+), 7 deletions(-)
/content
/content/radiogenomics_ml
```

```
1 from google.colab import drive
2 drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive")
```

Drive already mounted at /content/drive, to attempt to forcibly remount, call drive.mount('/content/drive',

```
1 !python3 run_analysis.py
```

```
--- [load_data] completato ---
```

```
Source richiesto : BOTH
```

```
Pazienti allineati: 54
```

```
Predittori totali : 857
```

```
Distribuzione :
```

```
Target
```

```
ADK      27
```

```
SCC      27
```

```
[neutral_feature_reduction] radiomica: 93 feature di texture/intensità, 14 feature di forma (ridotte separa
```

```
[variance_filter] 93 -> 93 feature (soglia=0.01)
```

```
[redundancy_reduction] 93 -> 36 feature (clustering su corr spearman, soglia=0.9)
```

```
[variance_filter] 14 -> 14 feature (soglia=0.01)
```

```
[redundancy_reduction] 14 -> 5 feature (clustering su corr spearman, soglia=0.9)
```

```
[select_genes] metodo=iqr_top_pct: 750 -> 37 geni
```

```
[redundancy_reduction] 37 -> 37 feature (clustering su corr spearman, soglia=0.9)
```

```
[neutral_feature_reduction] TOTALE: 857 -> 78 feature
```

```
=====
```

```
Modello: elastic_net
```

```
=====
```

```
[elastic_net] fold 1/5 | AUC=1.000 | BalAcc=0.917 | best_params={'clf__C': 0.01, 'clf__l1_ratio': 0.1}
```

```
[elastic_net] fold 2/5 | AUC=1.000 | BalAcc=1.000 | best_params={'clf__C': 0.01, 'clf__l1_ratio': 0.1}
```

```
[elastic_net] fold 3/5 | AUC=1.000 | BalAcc=0.900 | best_params={'clf__C': 0.05, 'clf__l1_ratio': 0.3}
```

```
[elastic_net] fold 4/5 | AUC=0.767 | BalAcc=0.733 | best_params={'clf__C': 1.0, 'clf__l1_ratio': 0.05}
```

```
[elastic_net] fold 5/5 | AUC=0.880 | BalAcc=0.900 | best_params={'clf__C': 0.01, 'clf__l1_ratio': 0.1}
```

```
[elastic_net] RISULTATI NESTED CV (media ± sd su 5 fold esterni)
```

```
auc: 0.929 ± 0.094
```

```
balanced_accuracy: 0.890 ± 0.087
```

```
f1: 0.887 ± 0.089
```

```
=====
```

```
Modello: random_forest
```

```
=====
```

```
[random_forest] fold 1/5 | AUC=0.867 | BalAcc=0.917 | best_params={'clf__max_depth': 3, 'clf__min_samples_l
```

```
[random_forest] fold 2/5 | AUC=1.000 | BalAcc=0.917 | best_params={'clf__max_depth': 3, 'clf__min_samples_l
[random_forest] fold 3/5 | AUC=0.967 | BalAcc=0.900 | best_params={'clf__max_depth': 3, 'clf__min_samples_l
[random_forest] fold 4/5 | AUC=0.833 | BalAcc=0.733 | best_params={'clf__max_depth': 3, 'clf__min_samples_l
[random_forest] fold 5/5 | AUC=0.840 | BalAcc=0.900 | best_params={'clf__max_depth': 3, 'clf__min_samples_l
```

```
[random_forest] RISULTATI NESTED CV (media  $\pm$  sd su 5 fold esterni)
```

```
auc: 0.901  $\pm$  0.069
```

```
balanced_accuracy: 0.873  $\pm$  0.070
```

```
f1: 0.869  $\pm$  0.071
```

```
=====
Modello: svm_linear
=====
```

```
[svm_linear] fold 1/5 | AUC=0.383 | BalAcc=0.500 | best_params={'clf__C': 0.001}
[svm_linear] fold 2/5 | AUC=1.000 | BalAcc=0.917 | best_params={'clf__C': 0.01}
[svm_linear] fold 3/5 | AUC=0.800 | BalAcc=0.733 | best_params={'clf__C': 0.01}
[svm_linear] fold 4/5 | AUC=0.800 | BalAcc=0.817 | best_params={'clf__C': 0.1}
[svm_linear] fold 5/5 | AUC=0.920 | BalAcc=0.800 | best_params={'clf__C': 0.1}
```

```
[svm_linear] RISULTATI NESTED CV (media  $\pm$  sd su 5 fold esterni)
```

```
auc: 0.781  $\pm$  0.213
```

```
balanced_accuracy: 0.753  $\pm$  0.140
```

```
1 import pandas as pd
2
3 def analizza_preprocessing(csv_pre_path: str, csv_post_path: str):
4     """
5     Legge i dati pre e post riduzione delle feature, divide tra genomica (gen)
6     e radiomica (rad) e analizza quali feature sono state eliminate o mantenute.
7     """
8     print("=== [ANALISI PREPROCESSING FEATURE] ===\n")
9
10    # 1. Caricamento file
11    try:
12        df_pre = pd.read_csv(csv_pre_path)
13        df_post = pd.read_csv(csv_post_path)
14    except FileNotFoundError as e:
15        print(f"Errore: Impossibile trovare il file. Dettagli: {e}")
16    return
```

```
10         return
11
12
13
14
15
16
17
18     # 2. Estrazione delle colonne e filtraggio per prefisso
19     feat_pre = {c for c in df_pre.columns if c.startswith('gen') or c.startswith('rad')}
20     feat_post = {c for c in df_post.columns if c.startswith('gen') or c.startswith('rad')}
21
22     # 3. Operazioni insiemistiche per trovare scartate e mantenute
23     feat_mantenute = feat_pre.intersection(feat_post)
24     feat_scartate = feat_pre - feat_post
25
26     # 4. Funzione di supporto per separare le modalità
27     def separa_modalita(features_set):
28         gen = {f for f in features_set if f.startswith('gen')}
29         rad = {f for f in features_set if f.startswith('rad')}
30         return gen, rad
31
32     # Divisione in blocchi
33     pre_gen, pre_rad = separa_modalita(feat_pre)
34     post_gen, post_rad = separa_modalita(feat_post)
35     scartate_gen, scartate_rad = separa_modalita(feat_scartate)
36
37     # 5. Stampa del Report Generale
38     print(f"Totale feature INIZIALI : {len(feat_pre)} (Gen: {len(pre_gen)} | Rad: {len(pre_rad)})")
39     print(f"Totale feature FINALI   : {len(feat_post)} (Gen: {len(post_gen)} | Rad: {len(post_rad)})")
40     print(f"Totale feature SCARTATE : {len(feat_scartate)}")
41     print("-" * 50)
42
43     # 6. Dettaglio scarti per modalità
44     if len(pre_gen) > 0:
45         print(f"Feature Genomiche eliminate: {len(scartate_gen)} su {len(pre_gen)} "
46               f"({(len(scartate_gen)/len(pre_gen))*100:.1f}% del blocco genomico)")
47     if len(scartate_gen) > 0:
48         # Mostra le prime 10 per non intasare il terminale, oppure salvale tutte su file
49         print(f"    Esempi gen eliminati: {sorted(list(scartate_gen))[:10]}")
50
51     if len(pre_rad) > 0:
52         print(f"\nFeature Radiomiche eliminate: {len(scartate_rad)} su {len(pre_rad)} "
```

```
53         f"({(len(scartate_rad)/len(pre_rad))*100:.1f}% del blocco radiomico)")
54     if len(scartate_rad) > 0:
55         print(f"     Esempi rad eliminati: {sorted(list(scartate_rad))[:10]}")
56
57     print("-" * 50)
58
59     # 7. Salvataggio liste complete per confronto agevole con il PDF
60     output_txt = "report_feature_ridotte.txt"
61     with open(output_txt, "w") as f:
62         f.write("=== FEATURE MANTENUTE ===\n")
63         f.write("\n".join(sorted(feet_mantenute)))
64         f.write("\n\n=== FEATURE SCARTATE ===\n")
65         f.write("\n".join(sorted(feet_scartate)))
66
67     print(f"\nLista completa delle feature mantenute e scartate salvata in: '{output_txt}'")
68
69 # CAMBIA INDIRIZZI COL CAMBIO SOURCE_DATA
70 csv_pre = "/content/drive/MyDrive/problem_solving/project/outputs/both/X_features.csv"
71 csv_post = "/content/drive/MyDrive/problem_solving/project/outputs/both/X_reduced_features.csv"
72 analizza_preprocessing(csv_pre, csv_post)
```

=== [ANALISI PREPROCESSING FEATURE] ===

Totale feature INIZIALI : 857 (Gen: 750 | Rad: 107)

Totale feature FINALI : 78 (Gen: 37 | Rad: 41)

Totale feature SCARTATE : 779

Feature Genomiche eliminate: 713 su 750 (95.1% del blocco genomico)

Esempi gen eliminati: ['gen__ACVR1C', 'gen__ADAM12', 'gen__ADGRE1', 'gen__ADM', 'gen__ADORA2A', 'gen__Ak

Feature Radiomiche eliminate: 66 su 107 (61.7% del blocco radiomico)

Esempi rad eliminati: ['rad__original_firstorder_10Percentile', 'rad__original_firstorder_Entropy', 'rad

Lista completa delle feature mantenute e scartate salvata in: 'report_feature_ridotte.txt'

```
1 !python3 run_diagnostics.py
```

```
--- [load_data] completato ---
```

```
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```

```
Pazienti allineati: 54
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[select_genes] metodo=iqr_top_pct: 750 -> 37 geni
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```
[redundancy_reduction] 37 -> 37 feature (clustering su corr spearman, soglia=0.9)
```

```
[neutral_feature_reduction] TOTALE: 857 -> 78 feature
```

```
[majority_vote_params] combinazione più frequente: {'clf__max_depth': 3, 'clf__min_samples_leaf': 5, 'clf__
```

```
[run_diagnostics] modello migliore (pooled OOF AUC): random_forest | iperparametri (voto di maggioranza): {
```

```
=====
```

TEST DI PERMUTAZIONE

```
=====
```

```
[permutation_test] AUC sui dati veri: 0.901
```

```
[permutation_test] 50/200 permutazioni completate
```

```
[permutation_test] 100/200 permutazioni completate
```

```
[permutation_test] 150/200 permutazioni completate
```

```
[permutation_test] 200/200 permutazioni completate
```

```
[permutation_test] AUC media con etichette casuali: 0.496 ± 0.104
```

```
[permutation_test] p-value empirico: 0.0050
```

```
[permutation_test] Il modello sui dati veri supera in modo significativo la distribuzione ottenuta per caso
```

```
[plot_permutation_test] salvato in /content/drive/MyDrive/problem_solving/project/outputs/both/permutation_
```

```
=====
```

CONTROLLO BATCH EFFECT

```
=====
```

```
[load_batch_column] Nessuna colonna di batch configurata (config.BATCH_COL è None): controllo di batch effe
```

```
[batch_effect_diagnostic] Nessuna variabile di batch fornita: controlla solo visivamente il pannello per fe
```

```
[batch_effect_diagnostic] plot salvato in /content/drive/MyDrive/problem_solving/project/outputs/both/batch_effect_diagnostic.png
Diagnostica completata. File salvati in: /content/drive/MyDrive/problem_solving/project/outputs/both
```

```
1 import os
2 print(os.cpu_count())
```

2

```
1 !python3 run_specification_curve.py
```

[illegible]

```
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'variance', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'variance', 'exclude_shape': False, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'variance', 'exclude_shape': False, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'variance', 'exclude_shape': False, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'variance', 'exclude_shape': False, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_pct', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_pct', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_pct', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_pct', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_pct', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_pct', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_pct', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_n', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_n', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_n', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_n', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_n', 'exclude_shape': False, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_n', 'exclude_shape': False, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_n', 'exclude_shape': False, 'specification_curve': True}
[specification_curve] {'data_source': 'genomics', 'gene_selection_method': 'iqr_top_n', 'exclude_shape': False, 'specification_curve': True}
[specification_curve] {'data_source': 'both', 'gene_selection_method': 'variance', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'both', 'gene_selection_method': 'variance', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'both', 'gene_selection_method': 'variance', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'both', 'gene_selection_method': 'variance', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'both', 'gene_selection_method': 'variance', 'exclude_shape': False, 'specification_curve': True}
[specification_curve] {'data_source': 'both', 'gene_selection_method': 'variance', 'exclude_shape': False, 'specification_curve': True}
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[specification_curve] {'data_source': 'both', 'gene_selection_method': 'iqr_top_pct', 'exclude_shape': True, 'specification_curve': True}
[specification_curve] {'data_source': 'both', 'gene_selection_method': 'iqr_top_pct', 'exclude_shape': True, 'specification_curve': True}
```

```
1 !python3 feature_consensus.py
```

```
[feature_consensus] modello migliore: random_forest (tree)
[feature_consensus] stability: /content/drive/MyDrive/problem_solving/project/outputs/both/stable_features_
[feature_consensus] shap: /content/drive/MyDrive/problem_solving/project/outputs/both/shap_feature_importance_
[feature_consensus] spec curve votes: /content/drive/MyDrive/problem_solving/project/outputs/feature_votes_
[feature_consensus] salvato in /content/drive/MyDrive/problem_solving/project/outputs/both/feature_consensus_
```

```
[feature_consensus] Top 15 feature per consenso:
```

```
[feature_consensus] top 15 feature per consenso:
```

	consensus_score	n_criteria_present
gen__NECTIN1	1.000000	3
gen__CD44	0.980655	3
gen__S100A8	0.976190	3
gen__HIF1A	0.970238	3
gen__EPCAM	0.953869	3
gen__COL17A1	0.950893	3
gen__S100A9	0.800595	2
gen__VCAN	0.787202	2
gen__TPI1	0.776786	2
gen__A2M	0.773810	2
gen__PKM	0.764881	2
rad__original_glszm_SizeZoneNonUniformity	0.758929	2
gen__CD74	0.747024	2
rad__original_firstorder_Skewness	0.741071	2
rad__original_gldm_DependenceEntropy	0.738095	2

```
1 # 1) config.DATA_SOURCE = "genomics" -> python run_analysis.py
2 # 2) config.DATA_SOURCE = "both"      -> python run_analysis.py
3 # 3)
4 !python3 compare_data_sources.py --model elastic_net --source_a genomics --source_b both
5 !python3 compare_data_sources.py --model random_forest --source_a genomics --source_b both
```

```
[DeLong] AUC genomics=0.907 | AUC both=0.877
```

```
[DeLong] differenza: -0.030 (SE=0.032) | 95% CI [-0.094, +0.033] | z=-0.93 | p=0.3511
```

```
[DeLong] -> CI include lo 0: nessuna evidenza che both migliori su genomics (test asintotico esatto, non
```

```
[compare_sources] modello=elastic_net | genomics vs both | n=54
```

```
AUC genomics: 0.907
```

```
AUC both: 0.877
```

```
differenza (both - genomics): -0.030 [95% CI -0.102, +0.030] | p=0.3630
```

```
-> il CI include lo 0: NON c'è evidenza sufficiente che both migliori su genomics con questo campione.
```

```
[plot_bootstrap_diff] salvato in /content/drive/MyDrive/problem_solving/project/outputs/comparisons/bootsr
```

```
[DeLong] AUC genomics=0.907 | AUC both=0.900
```

```
[DeLong] differenza: -0.007 (SE=0.024) | 95% CI [-0.054, +0.040] | z=-0.29 | p=0.7748
```

```
[DeLong] -> CI include lo 0: nessuna evidenza che both migliori su genomics (test asintotico esatto, non
```

```
[compare_sources] modello=random_forest | genomics vs both | n=54
  AUC genomics: 0.907
  AUC both: 0.900
  differenza (both - genomics): -0.007 [95% CI -0.063, +0.039] | p=0.8440
  -> il CI include lo 0: NON c'è evidenza sufficiente che both migliori su genomics con questo campione.
[plot_bootstrap_diff] salvato in /content/drive/MyDrive/problem_solving/project/outputs/comparisons/boots
```

```
1 !python3 radiogenomics.py
```

```
[load_stable_feature_sets] consensus letto da /content/drive/MyDrive/problem_solving/project/outputs/both/1
[load_stable_feature_sets] soglia n_criteria_present >= 2: 20 feature radiomiche, 15 geni stabili (su 113 f
```

```
=====
MATRICE DI CORRELAZIONE PER COPPIE (con correzione FDR)
=====
```

```
[pairwise_correlation_matrix] metodo=spearman | 300 coppie testate (20 rad x 15 geni) | 4 coppie con q<0.05
[plot_correlation_heatmap] salvato in /content/drive/MyDrive/problem_solving/project/outputs/radiogenomics/
```

Top 15 coppie per q-value:

	radiomic_feature	gene	correlation	p_value	q_value
	rad__original_glcml_JointEntropy	gen__TPI1	-0.543663	0.000022	0.006456
rad__original_gldm_SmallDependenceLowGrayLevelEmphasis		gen__TPI1	-0.513703	0.000071	0.010655
rad__original_gldm_LargeDependenceEmphasis		gen__TPI1	0.501734	0.000111	0.011103
rad__original_shape_MeshVolume		gen__TPI1	0.450200	0.000636	0.047704
rad__original_shape_SurfaceVolumeRatio		gen__TPI1	-0.431142	0.001135	0.068099
rad__original_gldm_SmallDependenceLowGrayLevelEmphasis		gen__A2M	0.396379	0.003005	0.071773
rad__original_glrlm_RunLengthNonUniformity		gen__TPI1	0.397370	0.002927	0.071773
rad__original_shape_MeshVolume		gen__PKM	0.405908	0.002325	0.071773
rad__original_gldm_SmallDependenceLowGrayLevelEmphasis		gen__PKM	-0.399276	0.002782	0.071773
rad__original_firstorder_Kurtosis		gen__TPI1	0.395083	0.003110	0.071773
rad__original_glrlm_RunLengthNonUniformity		gen__PKM	0.400114	0.002720	0.071773
rad__original_firstorder_Skewness		gen__TPI1	-0.411168	0.002011	0.071773
rad__original_glcml_JointEntropy		gen__PKM	-0.401487	0.002621	0.071773
rad__original_firstorder_Median		gen__TPI1	0.379150	0.004693	0.100564
rad__original_glcml_ClusterProminence		gen__A2M	0.373814	0.005363	0.102658

```
=====
COEFFICIENTE RV + TEST DI PERMUTAZIONE (associazione globale)
=====
```

```
[rv_permutation_test] n_pazienti=54 | n_feature_rad=20 | n_geni=15 | RV osservato = 0.1300
[rv_permutation_test] 5000 permutazioni dell'abbinamento paziente-gene...
[rv_permutation_test] 500/5000 permutazioni completate
[rv_permutation_test] 1000/5000 permutazioni completate
[rv_permutation_test] 1500/5000 permutazioni completate
[rv_permutation_test] 2000/5000 permutazioni completate
[rv_permutation_test] 2500/5000 permutazioni completate
[rv_permutation_test] 3000/5000 permutazioni completate
[rv_permutation_test] 3500/5000 permutazioni completate
[rv_permutation_test] 4000/5000 permutazioni completate
[rv_permutation_test] 4500/5000 permutazioni completate
[rv_permutation_test] 5000/5000 permutazioni completate

[rv_permutation_test] RV nullo (permutato): 0.0372 ± 0.0412
[rv_permutation_test] RV osservato: 0.1300
[rv_permutation_test] p-value empirico: 0.0364 (181/5000 permutazioni >= al reale) | CI 95% esatta (Clopper
[rv_permutation_test] Evidenza di associazione multivariata reale tra radiomica e genomica (feature stabili
[plot_rv_permutation_test] salvato in /content/drive/MyDrive/problem_solving/project/outputs/radiogenomics/

Tutti i risultati sono stati salvati in: /content/drive/MyDrive/problem_solving/project/outputs/radiogenomi
```

```
1 !python3 consensus_significance.py
```

```
[majority_vote_params] combinazione più frequente: {'clf__max_depth': 3, 'clf__min_samples_leaf': 5, 'clf__
[load_frozen_context] modello migliore: random_forest (tree) | iperparametri congelati: {'clf__max_depth':
[load_frozen_context] 72 combinazioni di specification curve congelate da specification_curve_results.csv

[permutation_null_consensus] 50 permutazioni, ognuna con: 200 bootstrap stability + 5-fold SHAP + l'intera
[Parallel(n_jobs=-1)]: Using backend LokyBackend with 2 concurrent workers.
[stability_selection] 50/200 bootstrap completati
[stability_selection] 50/200 bootstrap completati
[stability_selection] 100/200 bootstrap completati
[stability_selection] 100/200 bootstrap completati
[stability_selection] 150/200 bootstrap completati
[stability_selection] 150/200 bootstrap completati
[stability_selection] 200/200 bootstrap completati

[stability_selection] modello=random_forest | top_k=8 | 1 feature stabili (>= 60% dei bootstrap)
[stability_selection] 200/200 bootstrap completati
```

```

[stability_selection] 200/200 bootstrap completati

[stability_selection] modello=random_forest | top_k=8 | 0 feature stabili (>= 60% dei bootstrap)

[out_of_fold_shap] SHAP calcolata out-of-fold per 54 pazienti, 78 feature (model_type=tree)

[out_of_fold_shap] SHAP calcolata out-of-fold per 54 pazienti, 78 feature (model_type=tree)
[specification_curve] ATTENZIONE: modello collassato (nessuna feature selezionata in nessun fold) per spec=
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[specification_curve] ATTENZIONE: modello collassato (nessuna feature selezionata in nessun fold) per spec=

[specification_curve] 72 combinazioni (specificita x modello) testate | AUC pooled min=0.332 max=0.602

[Parallel(n_jobs=-1)]: Done    1 tasks      | elapsed:  5.2min

[specification_curve] 72 combinazioni (specificita x modello) testate | AUC pooled min=0.396 max=0.643

[stability_selection] 50/200 bootstrap completati
[stability_selection] 50/200 bootstrap completati
[stability_selection] 100/200 bootstrap completati
[stability_selection] 100/200 bootstrap completati
[stability_selection] 150/200 bootstrap completati
[stability_selection] 150/200 bootstrap completati
[stability_selection] 200/200 bootstrap completati

[stability_selection] modello=random_forest | top_k=8 | 0 feature stabili (>= 60% dei bootstrap)
[stability_selection] 200/200 bootstrap completati

[stability_selection] modello=random_forest | top_k=8 | 1 feature stabili (>= 60% dei bootstrap)

[out_of_fold_shap] SHAP calcolata out-of-fold per 54 pazienti, 78 feature (model_type=tree)

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[specification_curve] ATTENZIONE: modello collassato (nessuna feature selezionata in nessun fold) per spec=  
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```